Linux Administration Guide

Super User (Admin)

- In Linux, Admin is called as "super-user"
- Admin's login name is "root"
- Most modern Linux distributions disable "root" login for security
- To execute commands with admin privileges use "sudo" (if approved by system admin)
- Syntax: `sudo command`

```
sudo apt package_name
```

- `sudo` : stands for "superuser do"
- `apt` : stands for "Advanced Package Tool" (used to install, update, and remove software packages)
- `package_name` : name of the software package to be installed

Common Admin Commands

```
sudo apt update             # Update package lists for upgrades and new packages
sudo apt upgrade            # Upgrade all installed packages to their latest
versions
sudo apt install vim        # Install vim text editor
sudo apt install gcc        # Install GCC compiler
sudo apt install git        # Install Git version control system
sudo apt install python3    # Install Python 3 programming language
```

Directory Commands

Command	Description	Syntax
`pwd`	Print present working directory	`pwd`
`cd`	Change directory	`cd dirpath`
`ls`	List directory contents	`ls dirpath`
`mkdir`	Make directory	`mkdir dirpath`
`rmdir`	Remove empty directory	`rmdir dirpath`

Special CD Commands

- `cd ~` : Change to home directory

- `cd -` : Change to previous directory
- `cd /` : Change to root directory
- `cd .` : Stay in current directory
- `cd ..` : Change to parent directory

File Commands

Command	Description
`cat filepath`	Display file contents
`cat > filepath`	Create new file
`cat >> filepath`	Append to file
`touch filepath`	Create empty file Or Change file timestamp
`rm filepath`	Delete file
`rm -r dirpath`	Delete directory recursively
`mv filepath destpath`	Move/rename file
`cp filename newname`	Copy file
`cp -r dirpath destpath`	Copy directory recursively

File Types and Permissions

File Types

- Regular file (`-`)
- Directory (`d`)
- Link file (`l`)
- Pipe file (`p`)
- Socket file (`s`)
- Character special file (`c`)
- Block special file (`b`)

File Permissions

- `r` - read
- `w` - write
- `x` - execute
- Format: `(rwx)user (rwx)group (rwx)others`

Additional File Commands

Listing Commands

- `ls` - List files
- `ls` - list the contents of present working directory
- `ls path` - list the contents of given path

- `ls -l` - list the contents in detail format
  - e.g `$ls -l`
  - Output columns:
    - -rwxr-xr-x 1 user group 4096 Jan 01 12:00 filename
  - Columns represent:
    - File type
    - File permissions (rwx for user, group, others)
    - Number of links
    - Owner name
    - Group name
    - File size in bytes
    - Last modified date and time
    - File name

- options with `ls` command
  - `ls -h` - Human readable sizes
  - `ls -l` - Detailed listing
  - `ls -a` - Show hidden files
    - Hidden files are files that start with a dot (.) and are not displayed by default.
  - `ls -A` - Show hidden files (except . and ..)
  - `ls -i` - list contents with indoe number(inode number is unique number given to every file)
  - `ls -s` - list content with size (number of blocks)
  - `ls -S` - list content in descending order of their sizes
  - `ls -alh` - combine multiple options
  - `ls -R` - list contents recursively (including sub-directories)

- For more options use `man ls` command

File Information

- `stat file` - Display file information
- `head file` - Display first 10 lines
  - `head -n 5 file` - Display first 5 lines
  - `head -c 15 file` - Display first 15 bytes
  - `head -n 5 -c 15 file` - Display first 5 lines and first 15 bytes
- `tail file` - Display last 10 lines
  - `tail -n 5 file` - Display last 5 lines
  - `tail -c 15 file` - Display last 15 bytes
- `sort file` - Sort contents alphabetically
  - `sort -r file` - Sort contents in reverse order
  - `sort -n file` - Sort contents numerically
  - `sort -u file` - Sort contents and remove duplicates
  - `sort -k 2 file` - Sort contents based on second column
  - `sort -t , -k 2 file` - Sort contents based on second column with comma as delimiter
- `uniq file` - Display unique lines
  - `uniq -c file` - Display unique lines with count

    - `uniq -d file` - Display duplicate lines
    - `uniq -u file` - Display unique lines only
    - `uniq -i file` - Ignore case while comparing lines
    - `uniq -f 2 file` - Ignore first 2 fields while comparing lines
- `rev filepath` - Reverse each line
    - `rev -s filepath` - Reverse each line and sort
    - `rev -n filepath` - Reverse each line and remove duplicates
    - `rev -c filepath` - Reverse each line and count occurrences
- `tac filepath` - Reverse file contents
    - `tac -n filepath` - Reverse file contents and remove duplicates
    - `tac -c filepath` - Reverse file contents and count occurrences
- `wc file` - Word count (lines, words, characters)
    - `wc -l file` - Count lines
    - `wc -w file` - Count words
    - `wc -c file` - Count characters
- `file filepath` - Determine file type
- Use `man command` to get more information about any command.

Utility Commands

- `uname` - Display system information
    - `uname -a` - Display all system information
    - `uname -r` - Display kernel version
    - `uname -m` - Display machine hardware name
- `alias list="ls -l"` - Create command alias
- `unalias list` - Remove alias
- `which command` - Show command location
- `whereis command` - Show command and manual locations
- `date` - Display current date and time
- `cal` - Display calendar
- `clear` - Clear terminal screen
- `history` - Show command history
- `sleep seconds` - Pause for specified seconds
- `uptime` - Show system uptime
- `df` - Show disk space usage

Manual Pages

- man pages provide detailed information about commands, options, and usage.
- `man command` - Show manual page for a command
- Example: `man ls`
    - Use `q` to exit manual page viewer
    - Use `/searchterm` to search within the manual page
    - Use `n` to go to next search result
    - Use `Shift+n` to go to previous search result
- man pages are organized into sections:

1. User commands
2. System calls
3. Library functions
4. Special files
5. File formats
6. Games
7. Miscellaneous
8. System administration commands
